

# ASTRA clustering — project summary

context, methods, and design recommendations for the Fisher and the emulator

ASTRA clustering project

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## Abstract

A self-contained synthesis of the repository: the context, the underlying math, the ASTRA simulations, and — the core of this note — design recommendations (each backed by what we measured) for the **Fisher matrix** computation and the **full emulator**, followed by conclusions on what is most publishable and the recommended next steps. Companion technical notes carry the detail and figures: `notes/tier3_emulator/mlp_emulator_note.pdf` (emulator + inference) and `notes/fisher_emulator/fisher_emulator_note.pdf` (emulator-based Fisher).

## 1 Context

**ASTRA** classifies each galaxy by its local cosmic-web environment using a Delaunay tessellation over the combined data+random catalog, then measures the 2PCF *per environment quantile*. The scientific bet: splitting clustering by environment (voids  $\rightarrow$  knots) carries cosmological information *beyond the plain 2PCF*, because different environments respond differently to cosmology vs. to galaxy–halo (HOD) physics, breaking degeneracies. We pursue this on two tracks: a **Fisher forecast** (local derivatives  $\rightarrow$  forecast errors) and an **emulator + inference** pipeline (a forward model  $\xi(\theta)$  trained on simulations  $\rightarrow$  *measured* constraints via MCMC). A hard constraint throughout: **no new HOD generation** — we reuse existing HOD-populated boxes, single phase (ph000).

## 2 Basic math

**Environment classification.** From the data/random neighbour counts in the tessellation,

$$r = \frac{n_{\text{data}} - n_{\text{rand}}}{n_{\text{data}} + n_{\text{rand}}} \in [-1, 1],$$

binned into quantiles Q1 (void)...Q4 (knot). The 2PCF multipoles  $\xi_\ell(s)$  ( $\ell = 0$  monopole,  $\ell = 2$  quadrupole) are measured for each *leg*: full-sample auto, quantile autos, and full $\times$ quantile crosses, for data and for randoms.

**Fisher matrix.**

$$F_{ab} = \sum_{\text{bins}} \frac{\partial \xi}{\partial \theta_a} C^{-1} \frac{\partial \xi}{\partial \theta_b}, \quad \sigma(\theta_a) = \sqrt{(F^{-1})_{aa}}, \quad \text{FoM} = 1/\sqrt{\det F^{-1}}.$$

Its quality is set entirely by the *derivatives*  $\partial \xi / \partial \theta$  and the *covariance*  $C$ .

**Emulator + inference.** A neural emulator  $\mu(\theta) = \xi(\theta_{\text{cosmo}}, \theta_{\text{HOD}})$ ; Gaussian likelihood

$$\chi^2(\theta) = (d - \mu(\theta))^\top C_{\text{tot}}^{-1} (d - \mu(\theta)), \quad C_{\text{tot}} = C_{\text{CV}} + C_{\text{emu}} (+ C_{\text{label}}),$$

with  $C_{\text{CV}}$  the cosmic variance,  $C_{\text{emu}}$  the *emulator-error* covariance, and  $C_{\text{label}}$  the measurement (ASTRA-iteration) noise of the data vector; MCMC over  $\theta$  gives the posterior.

### Key diagnostics.

- **RMS/CV** — emulator prediction error in units of the cosmic-variance  $\sigma$ ;  $< 1$  is inference-grade.
- **value-add** =  $\sigma(2\text{PCF})/\sigma(2\text{PCF} + \text{ASTRA})$  — the tightening from environment splits.
- $C_{\text{emu}}/C_{\text{CV}}$  — emulator error relative to cosmic variance; sets whether a leg’s signal is usable in inference.

## 3 The ASTRA simulations

- **Boxes:** AbacusSummit,  $2 h^{-1} \text{ Mpc Gpc}$  per side (native), HOD-populated at  $z = 0.5$  with the yuan23 prior (`ntbfin` catalogs), *single phase* `ph000`;  $\sim 3\text{--}5 \text{ M galaxies/box}$ . Alcock–Paczynski rescaling to a common frame makes the effective side vary per cosmology ( $\sim 1.87\text{--}2.18 \text{ Gpc/h}$ , anisotropic);  $\xi(s)$  is measured in that AP frame.
- **RSD & AP (how they enter the runs):** both are applied at load time, from fields already in the catalogs. *RSD* — we take the line of sight along  $z$  and use the precomputed redshift-space coordinate `Z_RSD` (which carries the peculiar-velocity displacement  $v_z/aH$ ) for the  $z$ -axis while `X_PERP`, `Y_PERP` stay real-space; this is what populates the quadrupole. *AP* — we divide the transverse coordinates by the header factor `Q_PERP` ( $\propto$  the  $D_A$  ratio) and the line-of-sight by `Q_PAR` ( $\propto$  the  $H$  ratio), mapping each cosmology’s box into the common fiducial frame; this anisotropic rescaling is exactly why the effective box side differs per cosmology and is what makes  $w_0, w_a$  observable through the quadrupole.
- **Cosmologies:** (i) a *Fisher grid* — `c000` (Planck  $\Lambda\text{CDM}$ ) + neighbours varying the 4  $\Lambda\text{CDM}$  params ( $\omega_b, \omega_{\text{cdm}}, n_s, \sigma_8$ ) at  $\pm 2\text{--}3\%$ ; (ii) an *emulator block* `c130–c181` — the AbacusSummit emulator Latin hypercube varying the broader  $w_0 w_a \text{CDM+}$  set: 8 params ( $\omega_b, \omega_{\text{cdm}}, h, n_s, \alpha_s, N_{\text{ur}}, w_0, w_a$ );  $A_s$  fixed,  $\sigma_8$  derived.
- **HODs:** 500 draws/cosmology exist; we use  $\sim 20\text{--}100$  (maximin-selected).
- **Pipeline:** full-box (periodic BC, the emulator target) and  $500 h^{-1} \text{ Mpc}$  sub-boxes (the covariance, pooled  $\rightarrow \sim 576$  samples); each run averages  $N$  ASTRA iterations (independent random realisations;  $N = 3$  in the pilot). Data product:  $17 \text{ stems} \times 2 \text{ multipoles} \times 15 \text{ s-bins}$  ( $0\text{--}150 h^{-1} \text{ Mpc}$ ) per run.
- **Status:** all 52 emulator cosmologies ( $\times \sim 20\text{--}50$  HODs) and 9 Fisher cosmologies ( $\times 50$ ) are on disk (`c000` at 100); the coverage run filled the cosmology axis  $10 \rightarrow 52$ .

## 4 Recommendations — Fisher matrix design

### F1. Derivatives from the emulator at fixed HOD, not from the $\pm$ cosmology pairs.

*Justification:* the  $\pm 2\text{--}3\%$  pairs are not HOD-matched, so HOD mismatch contaminated  $\partial\xi/\partial\theta$  at the level of the cosmology signal — the  $\sigma_8$  derivative came out  $\sim 11\%$  of the expected  $2\xi$  and needed the Tier-0/1 noise-model+subtraction saga. The emulator gives  $\partial\xi/\partial\theta$  at *fixed* HOD (HOD-clean by construction) over all **8**  $w_0w_a$ CDM+ params (the  $\pm$  grid gave only the 4  $\Lambda$ CDM).

**F2. Use the realistic covariance  $C = C_{\text{CV}} + C_{\text{emu}}$ , not  $C_{\text{CV}}$  alone.** *Justification:* with  $C_{\text{CV}}$  only (perfect emulator) the Fisher crowns the environment crosses, but the value-add test showed that in real inference the environment-leg  $C_{\text{emu}}$  ( $\sim 3\text{--}7\times\text{CV}$ ) down-weights exactly those legs and the value-add evaporated ( $\times 1.0$ ). A  $C_{\text{CV}}$ -only Fisher is not predictive of achievable constraints.

**F3. Marginalise the HOD jointly, with emulator gradients for both blocks.** *Justification:*  $\partial\xi/\partial\theta_{\text{HOD}}$  from the same emulator gives a consistent (8+12)-param Fisher; the MCMC showed HOD-marginalisation inflates  $\Lambda$ CDM errors only  $\times 1.0\text{--}1.8$  ( $\omega_b, \omega_{\text{cdm}}$  HOD-robust;  $h, n_s$  mild) — far below the  $2\text{--}10\times$  the old gradient+prior approach feared.

**F4. Validate the Gaussian-linear Fisher against the MCMC; use DALI where it breaks.** *Justification:* the curated-vector MCMC recovers c000 unbiased and  $\sim$ Gaussian (Fisher reliable), but tier3/ $w_0w_a$  recoveries are biased  $2\text{--}4\sigma$  (non-Gaussian extrapolation) — Fisher  $\sigma$  must be cross-checked.

*Implementation:* `emufisher_{lib,build,forecast,campaign, validate}.py`; note `notes/fisher_emulator/`.

## 5 Recommendations — emulator design

### E1. Train with a covariance-weighted loss ( $1/C_{\text{CV}}$ ), not iteration-noise weighting.

*Justification:* the architecture sweep showed the iteration-noise loss *starved* the environment legs (cross-cosmology RMS/CV  $\sim 31$ ); CV-weighting cut that to  $\sim 2$  — a  $\sim 15\times$  accuracy gain on exactly the legs that gate the value-add. Per-leg networks added nothing beyond the weighting.

**E2. Curate the data vector; do not dump all legs.** *Justification:* the greedy ranks the **void $\times$ full-sample-cross monopole as the #1 leg**, ahead of the plain 2PCF. The curated vector (full 2PCF + void-random-cross monopole) gives a real value-add ( $\sigma \times 1.2\text{--}1.7$ ) and an unbiased c000 recovery ( $< 0.5\sigma$ ), whereas adding all six environment legs gave only  $\times 1.1$  (noisy legs dilute).

**E3. Cosmology coverage is the dominant lever.** *Justification:* filling the hull  $10\rightarrow 52$  cosmologies dropped the broad-prior monopole LOCO from  $\sim 18\times$  to  $\sim 3\times\text{CV}$ ; the residual failures and the tier3/ $w_0w_a$  recovery biases are 8-D **hull-corner extrapolation**, which only denser coverage fixes.

**E4. ASTRA iterations are a *minor* lever —  $N \approx 10$  is plenty, not 25.** *Justification:* a zero-compute test (propagating the stored per-iteration scatter as  $\sigma_1/\sqrt{N}$ ) found the **quadrupole iteration noise already  $\sim 0.2\text{--}0.4\times\text{CV}$  at  $N = 3$** , i.e.  $\sim 10\%$  of the quad emulator error — so more iterations barely move  $C_{\text{emu}}$ . (The earlier “quads need  $\sim 25$  iters” used noise/*spread*, not the inference-relevant noise/*CV*.)

**E5. HODs are saturated by  $\sim 30/\text{cosmology}$ .** *Justification:* the c000 HOD learning curve plateaus by  $K \approx 30$  and is flat to  $K = 70$ ; the HOD response is low-dimensional and pooled across cosmologies, and HOD-marginalisation is already cheap.

**E6. Campaign design: coverage-first,  $\sim 10$  iterations,  $\sim 30\text{--}40$  HODs** — *not* the original 120-HOD/high-iteration plan. *Justification:* E3–E5; at fixed compute the budget is far better spent on the cosmology axis (corners) than on saturated HODs or already-adequate iterations.

**E7. Monopoles are inference-grade now; quadrupoles ( $\rightarrow w_0, w_a$ ) are coverage-limited.** *Justification:* the full-2PCF + data/random monopoles emulate at  $\sim 1 \times \text{CV}$  near  $\Lambda\text{CDM}$  and the  $\Lambda\text{CDM}$  recovery is clean today;  $w_0/w_a$  rely on the quadrupole, whose bias is set by coverage.

## 6 What is most publishable

1. **The value-add is emulator-gated, not information-limited.** A general methodological result: an environment/marked statistic can carry real cosmological information (Fisher:  $1.3\text{--}2.4 \times / \text{param}$ ) yet add *nothing* in inference until its emulator error falls below cosmic variance. This reframes “does the statistic help?” as “can we emulate it accurately enough?”, with a quantitative target ( $C_{\text{emu}} \ll \text{CV}$ ).
2. **A simple curated ASTRA vector beats the 2PCF today** — full 2PCF + the single void  $\times$  full-cross monopole tightens  $\Lambda\text{CDM}$  by  $\times 1.2\text{--}1.7$  with an unbiased recovery and essentially free HOD marginalisation.
3. **The CV-weighted-loss fix** — a  $\sim 15 \times$  emulator-accuracy gain on the environment legs from matching the training loss to the inference metric; broadly useful for SBI emulators.
4. **The lever analysis** — coverage  $\gg$  iterations  $\approx$  HODs, each backed by a targeted test (LOCO learning curve, quad-noise  $1/\sqrt{N}$ , HOD plateau) — budget guidance for any simulation-based-inference campaign.
5. **An honest emulator-aware Fisher** —  $C_{\text{CV}} \rightarrow C_{\text{CV}} + C_{\text{emu}}$  and HOD-clean emulator derivatives, reconciling the optimistic Fisher with real MCMC.

**Most publishable single paper:** an emulator-based ASTRA methodology paper combining (1), (2), (3), with the lever analysis (4) as the campaign justification.

## 7 Recommended next steps

1. **Finish the emulator-based Fisher** (`emufisher_*`, running): the forecast  $\sigma$  table over data vectors, the Fisher-vs-MCMC validation, and the campaign FoM-vs- $C_{\text{emu}}$  curve.
2. **Run the corrected campaign:** all 52 c130–c181 cosmologies, *densify the hull corners*,  $\sim 10$  iterations,  $\sim 30\text{--}40$  HODs — driving the environment-leg  $C_{\text{emu}}$  toward  $\lesssim 0.1 \times \text{CV}$  (the greedy shows this unlocks  $\sim 5 \times$  FoM and the full value-add).
3. **A neighbourhood-matched  $C_{\text{emu}}$**  so broad-prior errors are not inflated by distant hull-corner cosmologies.
4. **An external, independent-phase mock suite** (e.g. EZmocks) for an honest covariance and unbiased real recoveries — the main residual weakness under no-new-HOD.
5. **Dark energy:** once coverage is dense, revisit  $w_0/w_a$  with the quadrupole and the higher-iteration runs — the only directions still coverage-limited.
6. **Production emulator:** an MLP (SUNBIRD-style) on the full campaign with the curated vector + CV-weighted loss; MCMC on held-out mocks; compare measured  $\sigma$  to this Fisher.